

# Biological Substances & Infectious Materials — Shipping Reference

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Purpose: Field-by-field reference for reviewing shipments of blood, blood products, diagnostic specimens, and infectious substances (the “bloodborne pathogens” lane). Original summary built from IATA DGR Section 6.3 / PI 620 & PI 650, ICAO TI, 49 CFR 173.134/173.199, and OSHA Bloodborne Pathogens Standard 29 CFR 1910.1030. Not a copy of any proprietary form.

## Step 1 — Classify the material correctly (this drives everything)

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- **Category A — UN2814** (infectious to humans) / **UN2900** (infectious to animals): capable of causing permanent disability or life-threatening/fatal disease. Proper shipping name: “Infectious substance, affecting humans” / “...affecting animals”. Packing Instruction **PI 620**.
- **Category B — UN3373**: an infectious substance not meeting Category A criteria. Proper shipping name: “Biological substance, Category B”. Packing Instruction **PI 650**. This is the common diagnostic-specimen category.
- **Exempt human/animal specimen**: minimal likelihood of pathogens present — marked “Exempt human specimen” / “Exempt animal specimen”, not regulated as DG (but still packed as triple packaging).
- **Genetically modified micro-organisms/organisms** — UN3245 where applicable.
- **Dry ice used as coolant** — still UN1845, declared separately (see [cold-chain/03-dry-ice/](#)).

## Step 2 — Triple packaging (required for A, B, and exempt)

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- Primary receptacle (leak-proof, watertight) — for liquids, absorbent material between primary and secondary sufficient to absorb the entire contents.
- Secondary packaging (leak-proof, watertight).
- Rigid outer packaging with one surface  $\geq 100 \text{ mm} \times 100 \text{ mm}$ .
- Category A must use packaging certified to **UN Class 6.2 / PI 620 (P620)** and pass the drop test.
- Category B (PI 650): at least one surface of the outer package must display the UN3373 mark (diamond, side  $\geq 50 \text{ mm}$ ); proper shipping name “Biological substance, Category B” next to it.

## Step 3 — Marks, labels, documentation

| Item                        | Category A (UN2814/2900)               | Category B (UN3373)                        |
|-----------------------------|----------------------------------------|--------------------------------------------|
| Hazard label                | Class 6.2 Infectious Substance diamond | No 6.2 diamond; UN3373 mark only           |
| UN mark                     | UN2814 / UN2900 next to PSN            | UN3373 diamond mark                        |
| Shipper's Declaration (DGD) | <b>Required</b>                        | <b>Not required</b>                        |
| Air Waybill nature of goods | PSN + UN number                        | "Biological substance, Category B, UN3373" |
| Itemized list of contents   | Required (inside secondary)            | Recommended                                |
| 24-hr emergency contact     | Required                               | Not required by DGR, good practice         |
| Responsible person + phone  | Required                               | Required on the outer package              |

## Step 4 — Cold chain / OSHA overlap

- If shipped chilled/frozen with dry ice: add UN1845, net dry ice weight in **kg**, Class 9 handling, vented packaging ( `cold-chain/03-dry-ice/` ).
- Temperature-sensitive biologicals also carry a temp log — audit against `pharma/01-gdp-gmp/` .
- OSHA 29 CFR 1910.1030 governs the handling side (labeling of blood/OPIM containers as biohazard, handler training) — flag if internal handling documentation is absent.

## Review / practice checklist

- ☐ Classification correct (A vs B vs exempt) — the single most important call
- ☐ Correct UN number and proper shipping name for the classification
- ☐ Category A: DGD present, signed, PI 620 packaging certified; Category B: NO DGD, UN3373 mark present
- ☐ Triple packaging described; absorbent present for liquids
- ☐ Outer package  $\geq 100 \times 100$  mm; UN3373 mark  $\geq 50$  mm side (Cat B)
- ☐ Responsible person name + 24-hr phone on the package
- ☐ Dry ice (if used) declared as UN1845 with net kg and Class 9 handling
- ☐ Shipper/consignee consistent across AWB, DGD (if any), and packing list
- ☐ Import permit / sender's permit present where the destination requires one

## Common findings

-  Category A shipment tendered without a Shipper's Declaration — Critical.

-  Infectious substance mis-classified as UN3373 when it meets Category A criteria — Critical.
-  No absorbent material for a liquid primary receptacle — Critical (leak risk).
-  UN3373 mark under 50 mm or missing proper shipping name beside it — Major.
-  Responsible-person phone number missing from outer package — Major.
-  Itemized content list omitted on a Category B shipment — Minor.

## Sources

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- IATA DGR Infectious Substances program page: <https://www.iata.org/en/programs/cargo/dgr/infectious-substances/>
- 49 CFR 173.134 (Class 6.2 definitions): <https://www.ecfr.gov/current/title-49/subtitle-B/chapter-I/subchapter-C/part-173/subpart-D/section-173.134>
- OSHA Bloodborne Pathogens 29 CFR 1910.1030: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1030>
- WHO Guidance on regulations for the transport of infectious substances: <https://www.who.int/publications/i/item/9789240010086>